

Sepehr Abdous

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 sepehrabdous.github.io

Researcher in Datacenter Networking and AI Infrastructure

Education

Johns Hopkins University

PhD

School of Computer Science

Baltimore

2019-2026

Johns Hopkins University

M.Eng, GPA: 3.81/4

School of Computer Science

Baltimore

2019-2023

Sharif University of Technology

BSc, GPA: 17.06/20

School of Computer Engineering

Tehran

2014 - 2019

Research Summary

During my Ph.D., advised by Dr. Soudeh Ghorbani, my research focused on improving the performance and reliability of modern networks. I developed mechanisms to **detect and quantify congestion events** in **datacenter networks**, the critical infrastructure underpinning much of today's Internet. I also proposed techniques to **mitigate congestion within individual datacenters and across interconnected datacenter fabrics**. Later in my Ph.D., as large-scale Artificial Intelligence (AI) training workloads rapidly expanded, **my research shifted toward emerging challenges in AI datacenters**, including reducing bandwidth inefficiencies in distributed training jobs and enabling fast AI training by designing scalable and performant networking solutions.

Work Experience

Microsoft

Part-time consulting researcher

Under the supervision of Dr. Kabbani and Dr. Ghalayini, I designed Uno, a unified congestion control scheme that operates both within and across large-scale datacenter networks.

2024-2025

Baltimore, USA

CCC Intelligent Solutions

Intern

As an intern on the AI Enablement team, I designed and implemented a high-performance data migration technique that significantly improved the speed of transfers between the company's remote storage servers.

2022

Chicago, USA

Johns Hopkins University

Teacher Assistant

2019-2025

Baltimore, USA

During my Ph.D., I served as a TA 3 times across the following courses: “Computer Networks” and “Cloud Computing”.

HPDS (High Performance Data Storage) Company

2018

Intern

Tehran, Iran

I was responsible for automating the system update process and performing system-level FIO benchmarking tests.

Darmaneh

2016-2018

Frontend developer

Tehran, Iran

Darmaneh was a startup focused on building a platform for online medical consultations. I was responsible for developing and maintaining the Android and iOS applications.

Sharif University of Technology

2015-2019

Teacher Assistant

Tehran, Iran

As an undergraduate, I served as a teaching assistant 14 times across the following courses: “Fundamentals of Programming”, “Advanced Programming”, “Data Structures”, “Operating Systems”, “Numerical Methods”, “Computer Structure”, and “Mathematical Statistics”.

Skills

- **Technical Skills:** Datacenter networking, AI networking, Network simulation and hardware implementation
- **Programming Languages:** C, C++, Python, Java, P4
- **Tools:** OMNeT++, HTSim, Git, Latex, Ansible

Publications

Papers

One to Many: Closing the Bandwidth Gap in AI Datacenters with Scalable Multicast **2025**
Sepehr Abdous, Jinqi Lu, Jiacheng Wan, Erfan Sharafzadeh, Ying Zhang, Soudeh Ghorbani
HotNets

Uno: A One-Stop Solution for Inter- and Intra-Datacenter Congestion Control and Reliable Connectivity **2025**
Tommaso Bonato*, **Sepehr Abdous***, Abdul Kabbani, Ahmad Ghalayini, Nadeen Gebara, Terry Lam, Anup Agarwal, Tiancheng Chen, Zhuolong Yu, Konstantin Taranov, Mahmoud Elhaddad, Daniele De Sensi, Soudeh Ghorbani, Torsten Hoeﬂer
(*Equal contribution)
Codebase: https://github.com/spcl/Uno_SC25.git
SC

Tempus: Probabilistic Network Latency Verification 2024
Sepehr Abdous*, Senapati Diwangkara*, Soudeh Ghorbani
(*Equal contribution)
Codebase: <https://github.com/hopnets/Tempus>
[IEEE Access](#)

Practical Packet Deflection in Datacenters 2023
Sepehr Abdous, Erfan Sharafzadeh, Soudeh Ghorbani
Codebase: https://github.com/hopnets/practical_deflection
[CoNEXT](#)

Understanding the impact of the host networking elements on traffic bursts 2023
Erfan Sharafzadeh, **Sepehr Abdous**, Soudeh Ghorbani
Codebase: <https://github.com/hopnets/valinor-artifacts>
[NSDI](#)

Burst-tolerant Datacenter Networks with Vertigo 2021
Sepehr Abdous*, Erfan Sharafzadeh*, Soudeh Ghorbani
(*Equal contribution)
Codebase: <https://github.com/hopnets/vertigo-artifacts>
[CoNEXT](#)

Posters.....

Implementable selective deflection with Canary 2023
Sepehr Abdous, Erfan Sharafzadeh, Soudeh Ghorbani
[NSDI](#)

Vertigo: A Priority-aware Burst-tolerant Datacenter Fabric 2021
Sepehr Abdous, Erfan Sharafzadeh, Sougol Gheissi, Soudeh Ghorbani
[Coordinated Science Laboratory Student Conference \(CSLSC\)](#)

Valinor: Transport-Agnostic Packet Prioritization and Ordering at the Edge 2021
Erfan Sharafzadeh, **Sepehr Abdous**, Sougol Gheissi, Soudeh Ghorbani
[Coordinated Science Laboratory Student Conference \(CSLSC\)](#)

Professional Service

- Journal reviewer for Computer Networks (2022-2024)
- Journal reviewer for Journal of Network and Computer Applications (2022-2024)
- Journal reviewer for Ad Hoc Networks (2022-2023)